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Cryptology Assignment-4

Rail Fence

1) Encrypt the message "Enjoy your Journy" using Rail Fence with depth 2 & 3.

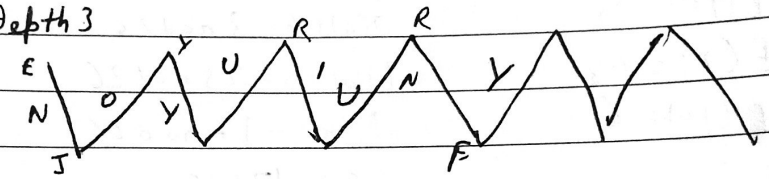
1) Encrypt Depth 2

E J Y O R O R E

N O Y U J U N Y

Ciphers: EJYORORE NOYUJUNY

Depth 3



Cipher: EYRR NOYUJUNY JOOE

2) Encrypt the message as per your choice using Rail Fence with depth 2&3

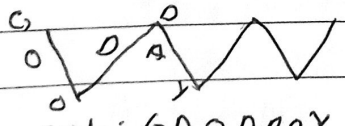
Depth 2 Plain Text: Good day

G O D Y

O D A

Cipher: GODYODA

Depth 3

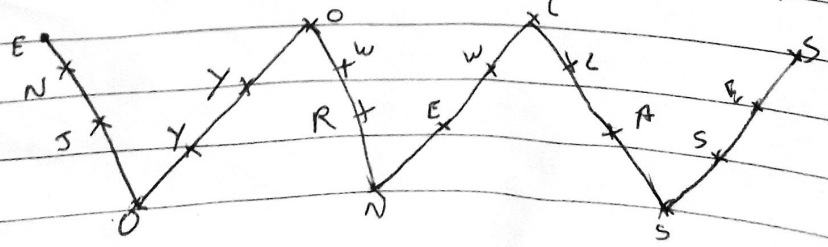


Cipher: GDODAOY

3) Decrypt the Cipher "EOCSNYUWLEJYREASONS", encrypted using Rail fence with depth 4

Depth 4 Length: 19

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19



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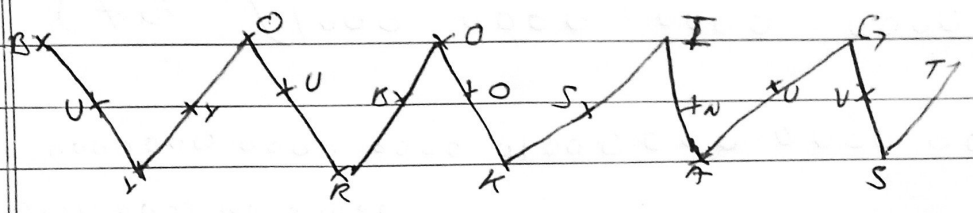
plaintext: ENJOY YOUR NEW CLASSES

4 Decrypt the ciphertext "BOOI GU YUBOSNUUTYRKAS", encrypted using RAIL fence with depth 3

Sol Depth: 3 length: 20

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

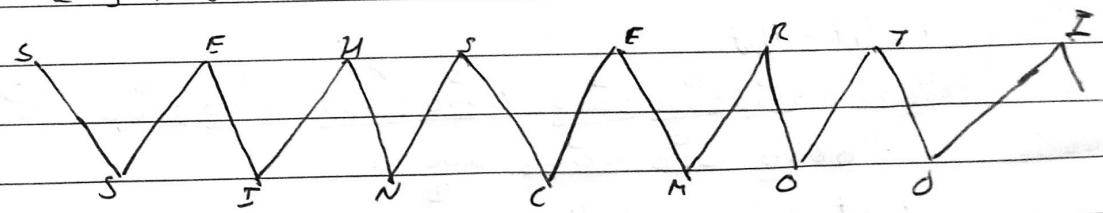


Plaintext - BUY YOUR BOOKS IN AUGUST

5 Decrypt the Ciphertext "SEMSERTISINCMOO" encrypted using Rail fence with depth 2.

Ans Depth: 2 length: 16

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



Plaintext: SSE I MNSCEMROTOI

Permutation

1 Encryption Encrypt the following message "we are discovered" using the key "53124"

5 3 1 2 4

W E A R E

D I S C O

V E R E D

Ciphertext: ASRRCEEIEEODWDV

DES

1 Using the initial permutation table, determine the effect output of

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the initial permutation box when the input is given in hexadecimal as: 0x0002 0000 0000 0001

Sol) Hexadecimal: 0x0002 0000 0000 0001

Binary: 0000 0000 0000 0010
0000 0000 0000 0000
0000 0000 0000 0000
0000 0000 0000 0001
} input to box (plain text)

0000 0000 0000 0010 0000 0000 0000 0000
0000 0000 0000 0000
0000 0000

15th
0000 0000 0000 0010 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0001

001A
16th
Input bit 64=1, output position 25=1
Input bit 15=1, output position 63=1
} By using permutation

64-bit output
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0010
2
output in hexadecimal:
0x0000 8000 0000 0002

2) Given the 4 bit Plaintext (M) 1101 and key (K) 1011, calculate the resulting ciphertext (C) using a bitwise Exclusive-OR (xor) operation. Show your bit-by-bit work

Sol) Plaintext: 1101
Key: 1011

1101
1011
0110

XOR

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

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Ciphertext: 0110

3) The input to S-box 1 is 000000. What is the output

000000
1st 6th bit

1st & 6th bit → row
Middle 4 bit → column

So, 0th Row & 0th column set the number in S-box-1 is

Binary conversion of 14 → 1110

Output = 1110